
CS 581

Introduction to Health Information Systems

HIS Population Health & Revenue Cycle Functionality

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Learning Objectives for this Live Classroom



Overview of HIS Functionality



What is Population Health & the HIS Functionality to Support it.

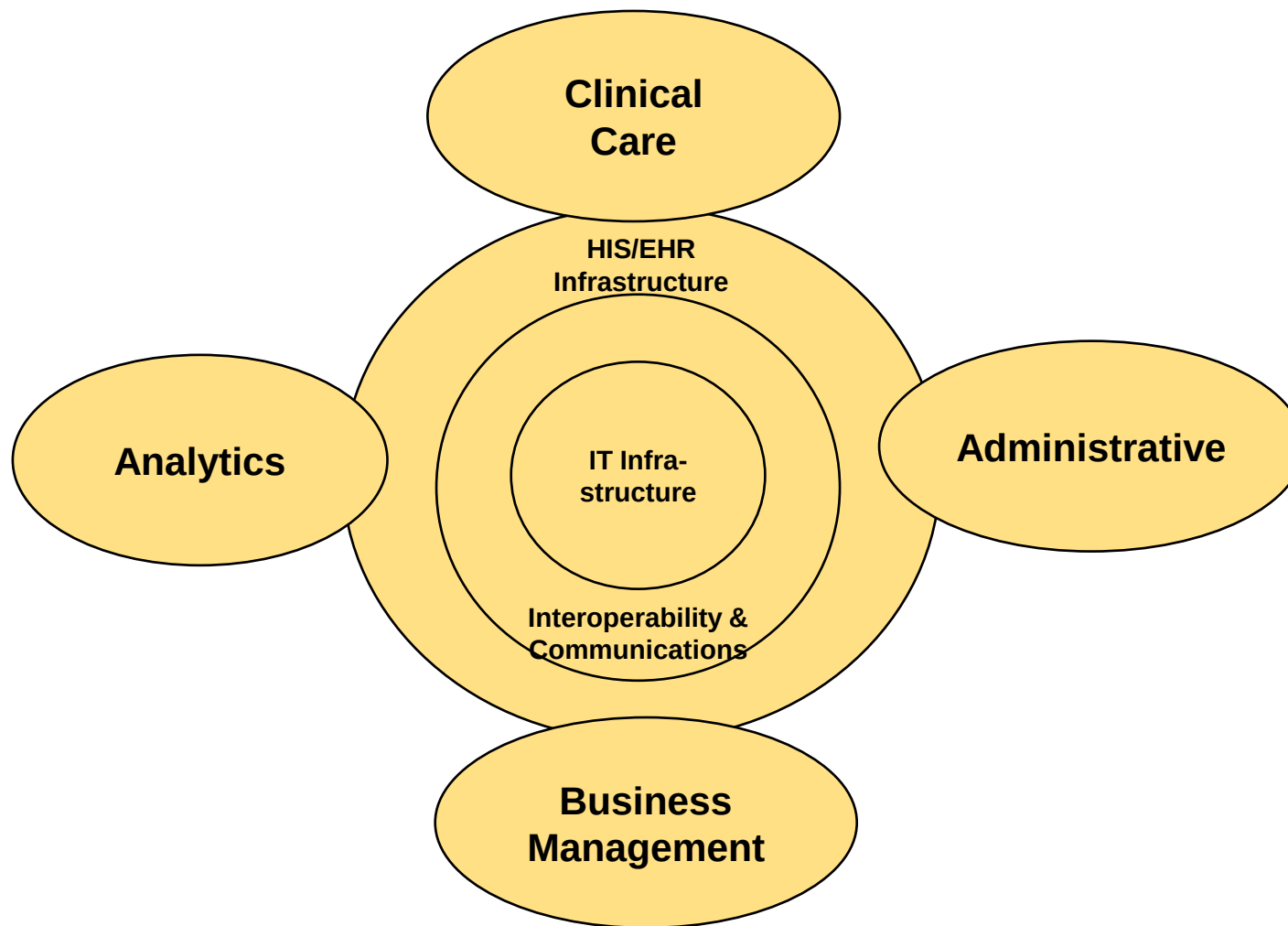


What is Revenue Cycle Management & the HIS Functionality to Support it.

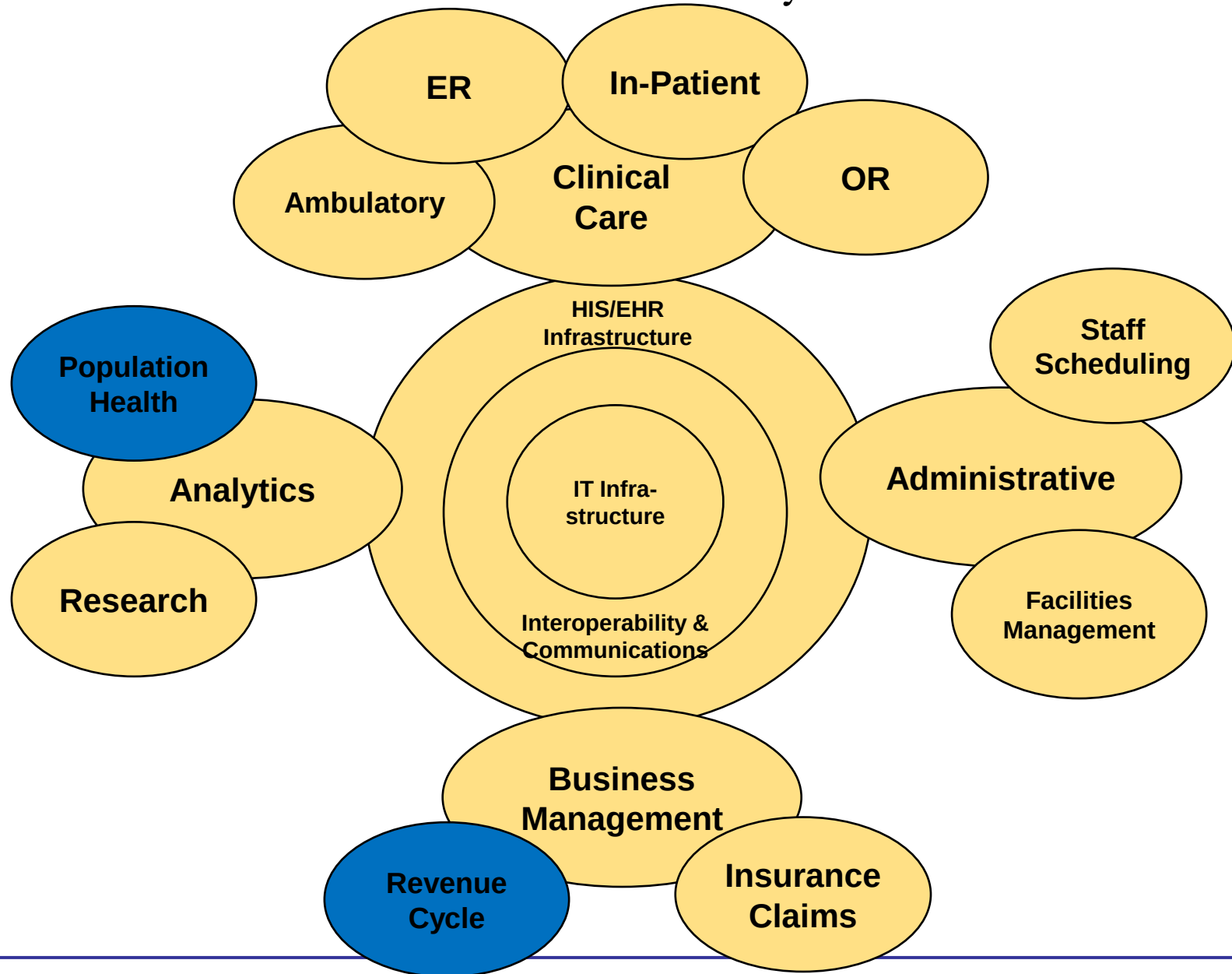


Overview of Assignment 3

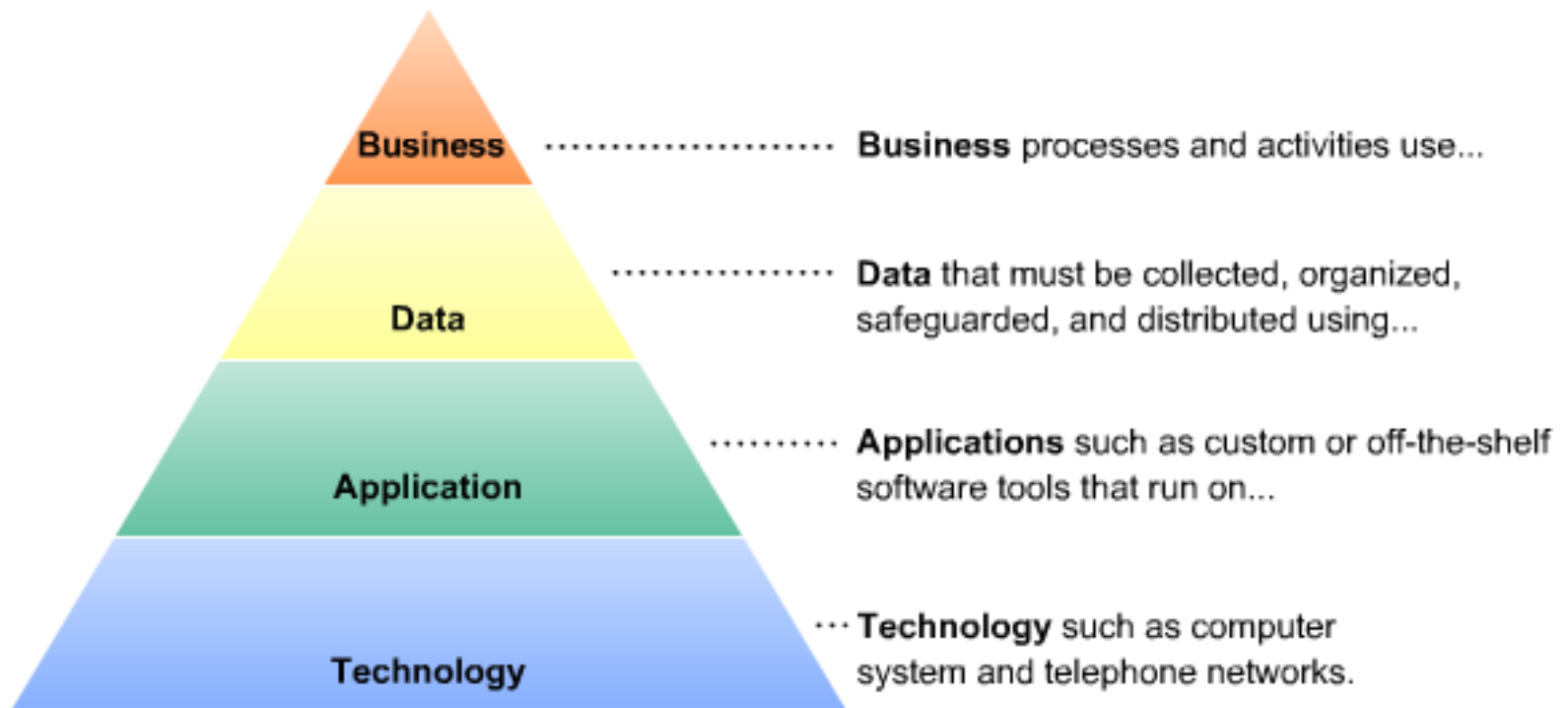
HIS/EHR Overall Functionality



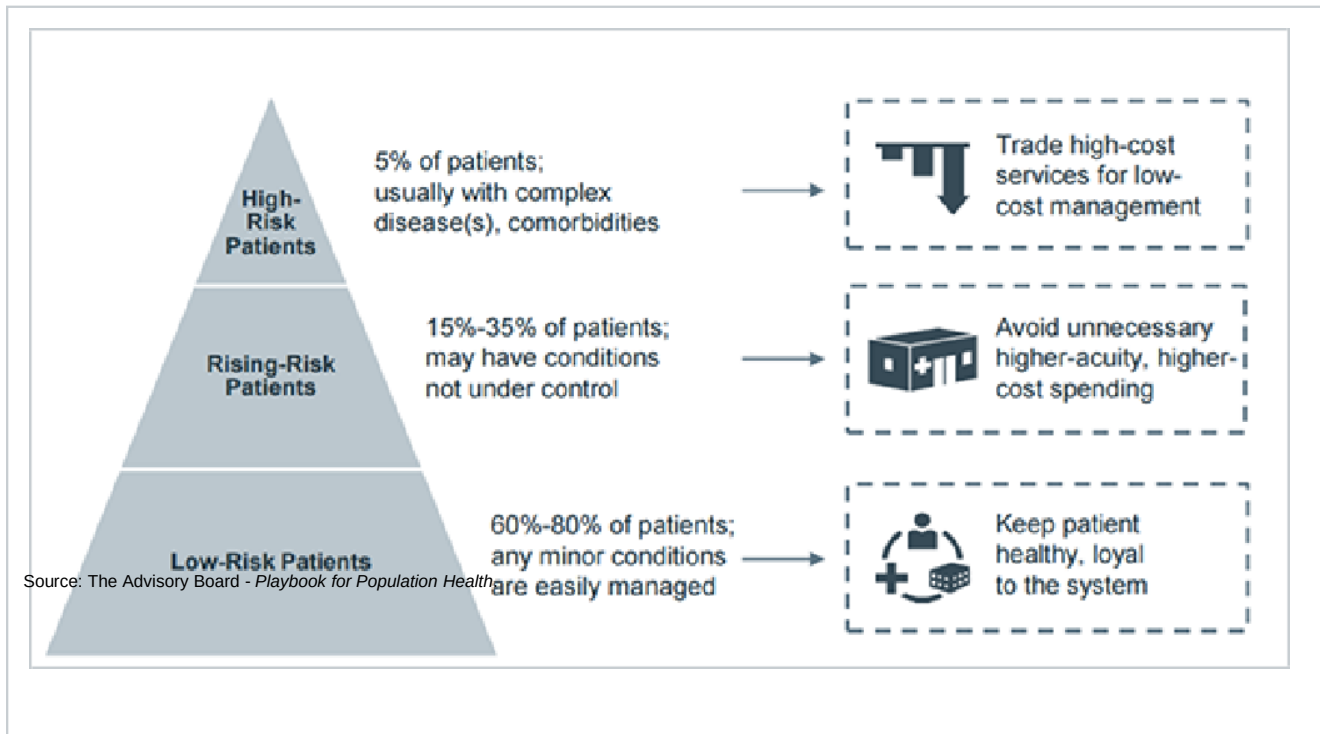
HIS/EHR Overall Functionality We Cover Today



Enterprise Architecture Overview



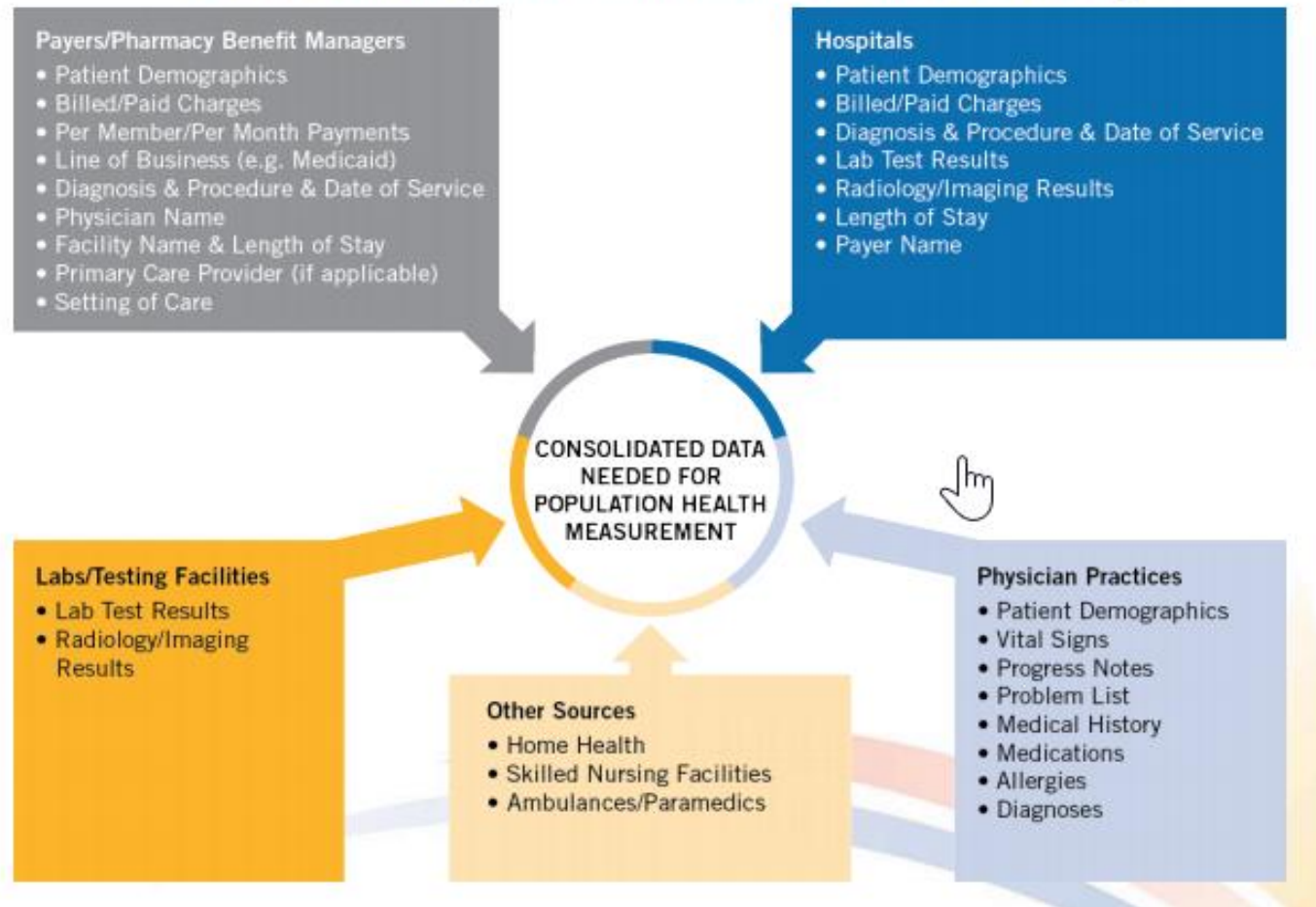
Population Health Business Problem



- Optimize the Triple Aim
 - Quality, Patient Experience & Cost
- Growing importance under managed care
- Impacts all parts of workflow!

Population Health Data Needed

Exhibit 1: Common Population Health Measurement Data Sources and Types



Source: <https://www.healthit.gov/sites/default/files/onc-beacon-lg6-it-for-pop-hlth-measmt.pdf>

? What Applications are Needed for
Population Health?

Population Health - Applications

MANY Application Systems – Including:

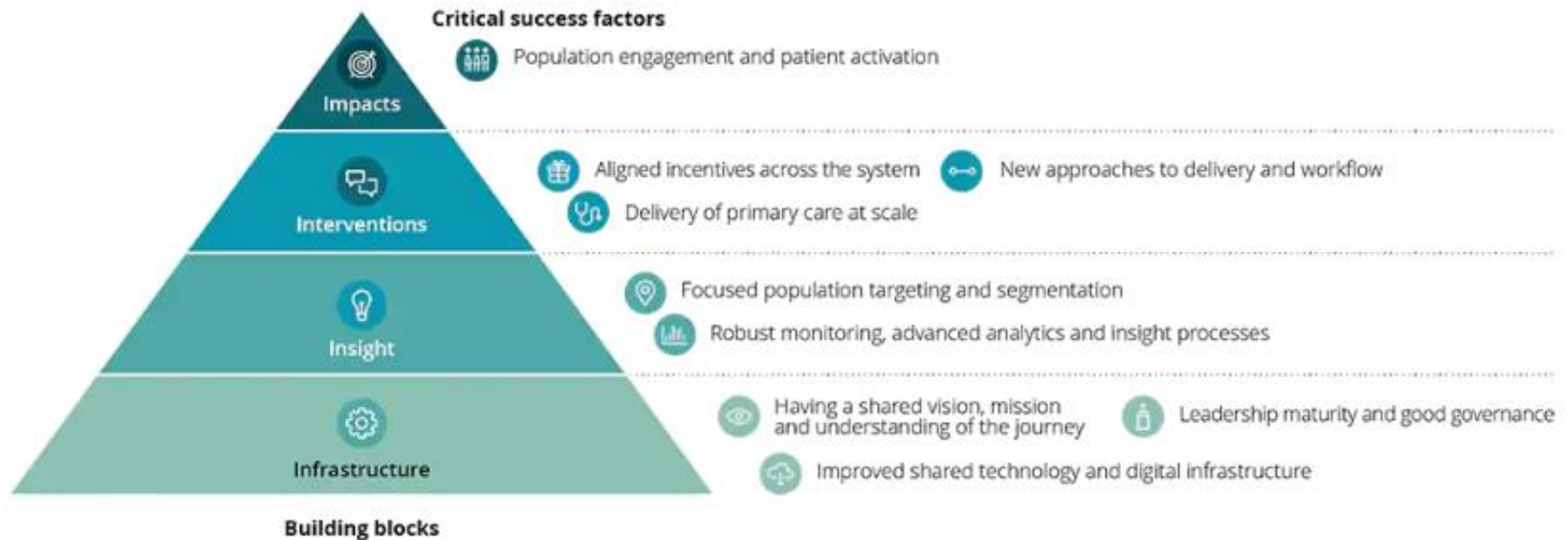
- Embedded in HIS and EHR systems
 - e.g. Epic, Cerner
- Standalone platforms and applications
 - Athenahealth Population Health, Health Catalyst, Innovaccer, Optum Population Health, Philips Wellcentive
- Healthcare data integration tools
 - e.g. HIEs, Athenahealth, Advisory Board
- “Big Data”
 - Data warehouse systems, R
- Using “traditional” data analytic tools
 - e.g. SASS, SAP
- “New technology” data analytics tools & AI
 - Incorporated AI functionality in an HIS
 - Athenahealth Population Health, Health Catalyst, IBM Watson Health, Innovaccer, Jvion, Lumiata, Microsoft Azure AI for Healthcare, Optum IQ,

Population Critical Success Factors

What do you – as a technologist - think
is critical to successfully enable
population health?

Population Health Critical Success Factors

The four key building blocks – Infrastructure, Insights, Impact and Intervention – and nine critical success factors enabling population health management



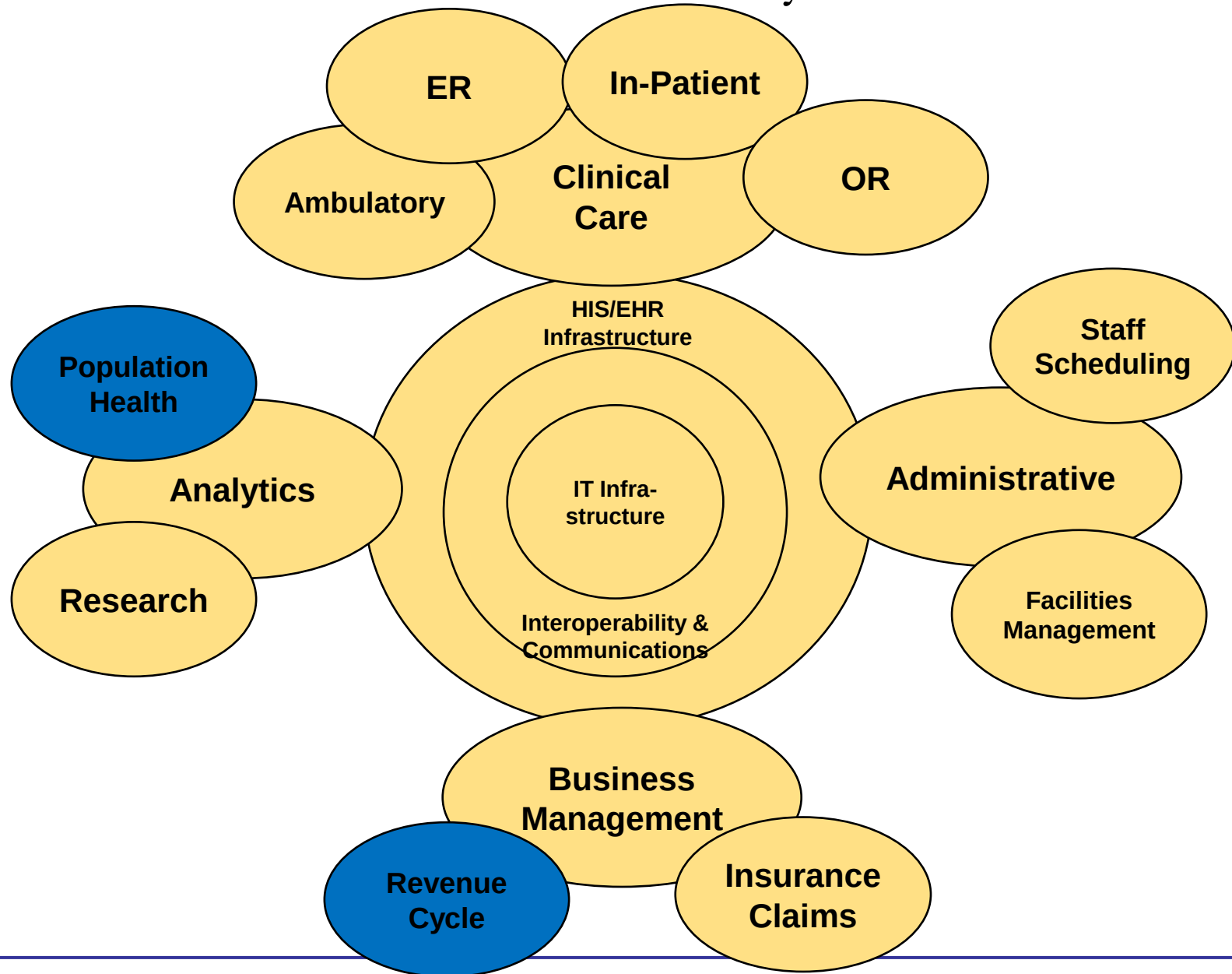
Source: Deloitte <https://www2.deloitte.com/uk/en/pages/public-sector/articles/enabling-population-health.html>

Population Health Critical Success Factors

- 1. Build collaboration, consensus, and commitments among key stakeholders.**
- 2. Identify and engage data sources and owners to obtain access to required data.**
- 3. Design and implement data access, transmission, and analytics processes.**
- 4. Continuously monitor and improve data quality.**
- 5. Develop and implement reporting on population health measures.**

Source: <https://www.healthit.gov/playbook/population-public-health/#section-10-3>

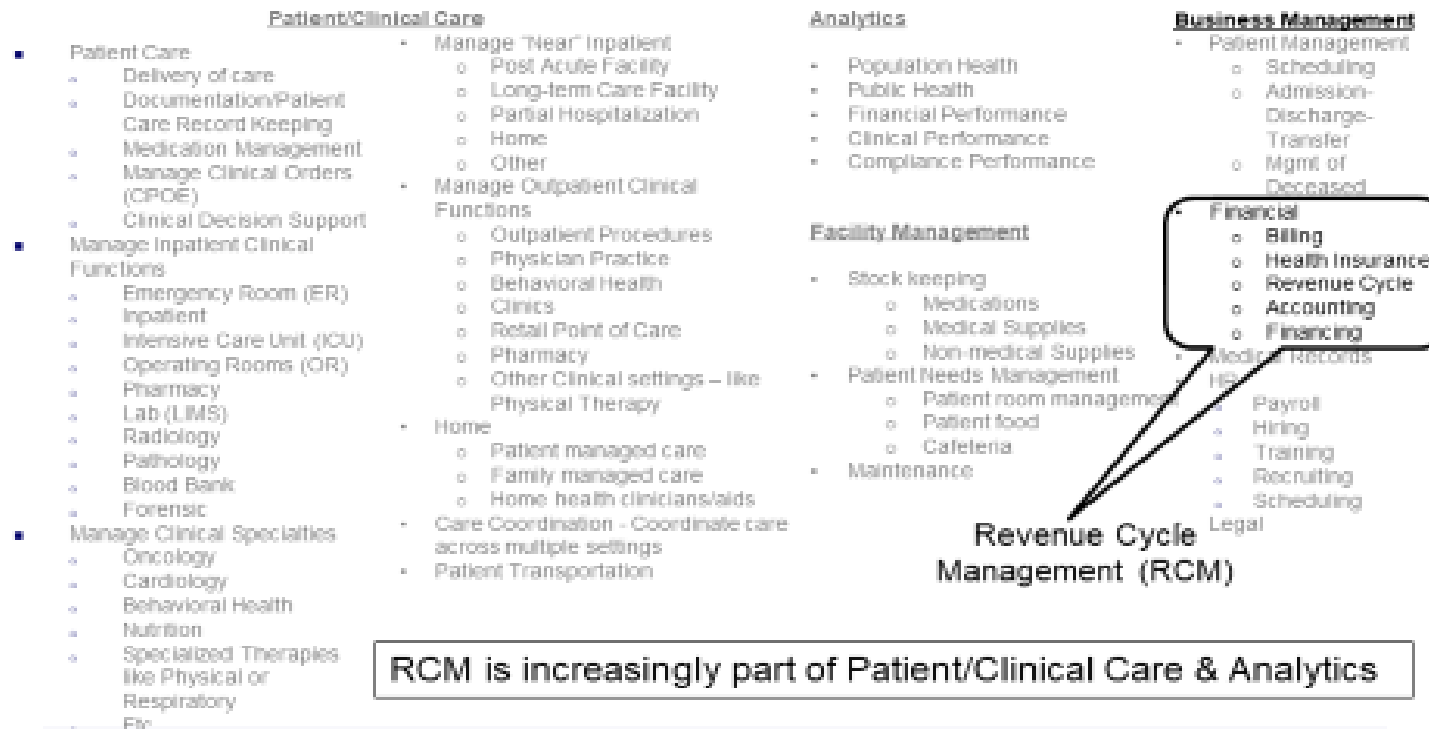
HIS/EHR Overall Functionality We Cover Today



Revenue Cycle Management – Business Functions

Business Layer

What Are The “Business” Functions of Healthcare?



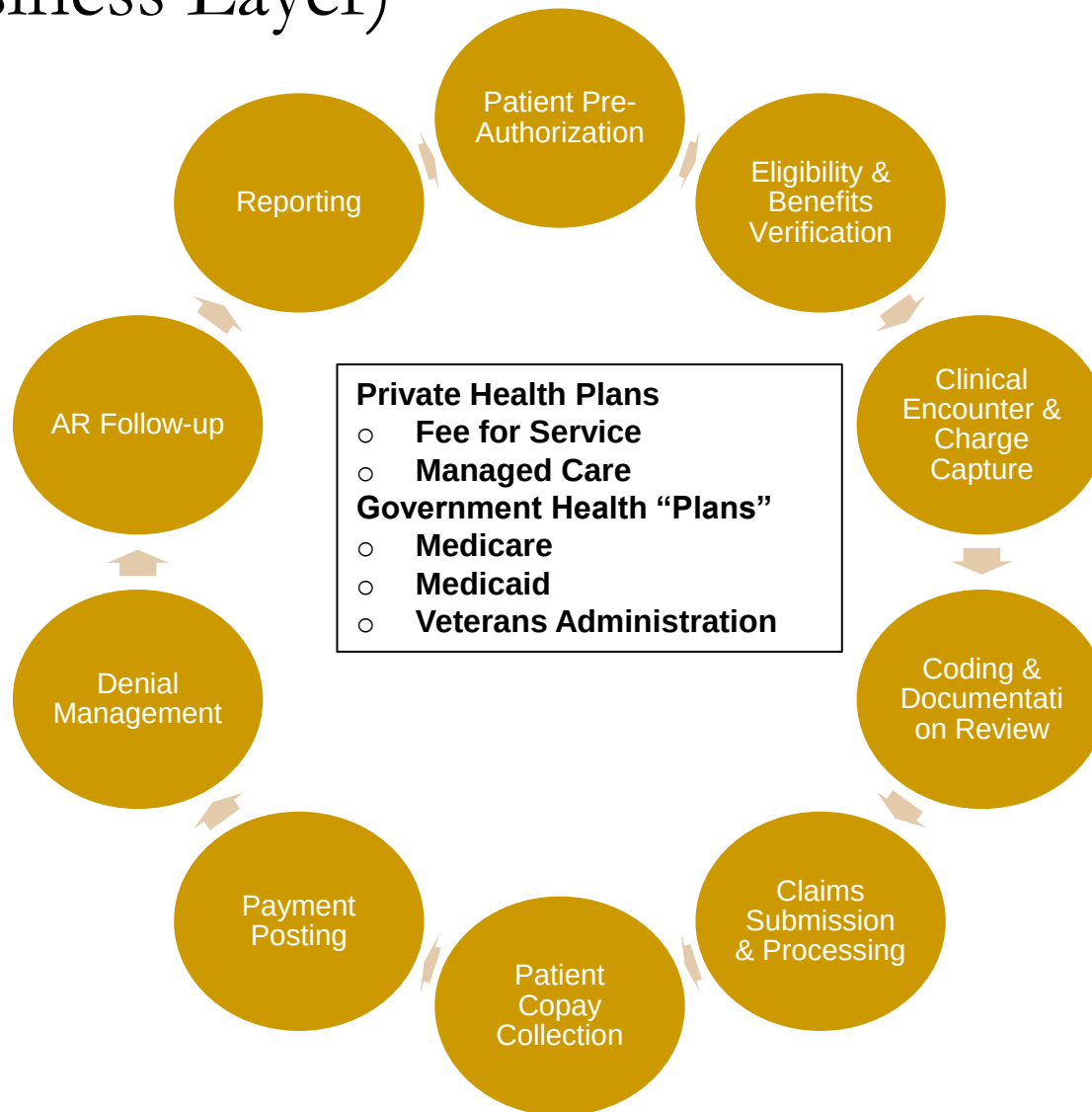
Revenue Cycle Management – Business

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What is Revenue Cycle
Management?

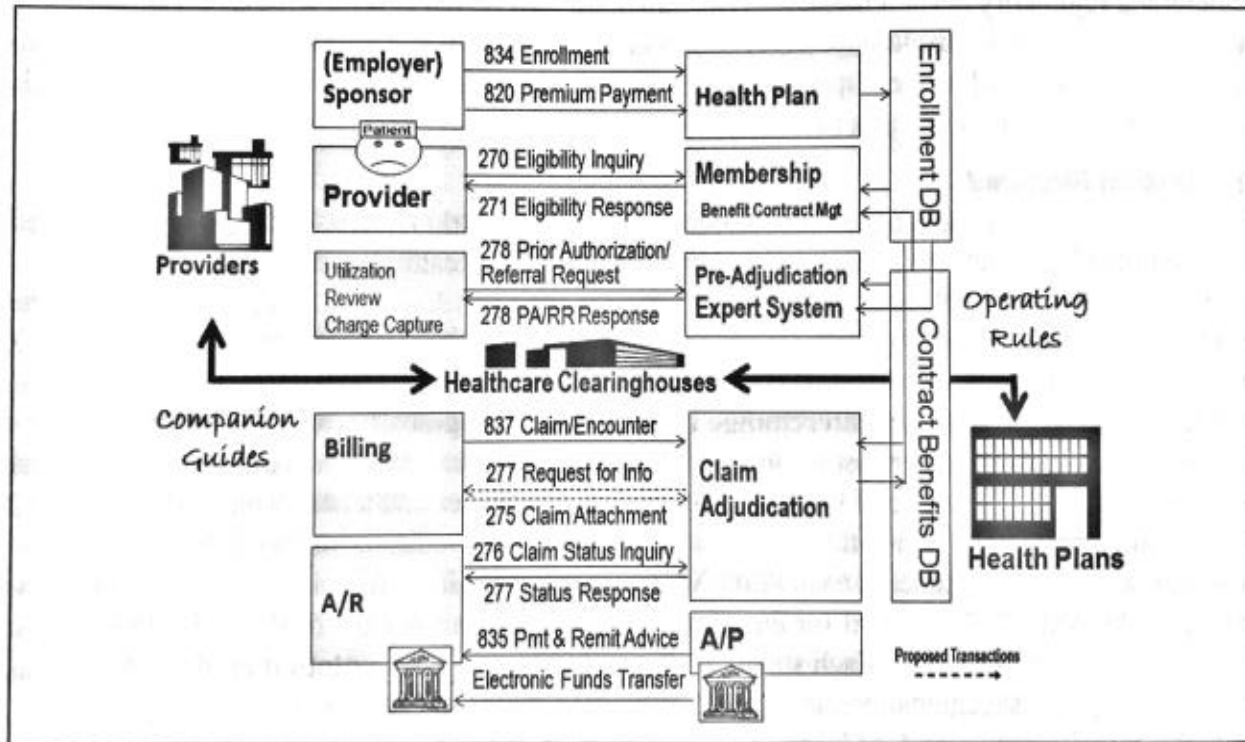
What business functions does
RCM include?

Revenue Cycle Management – Workflow (EA Business Layer)



Revenue Cycle Management – EA Data Layer

Figure 19.1. Flow of financial and administrative transactions



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- Complex data from multiple sources
- Integration with payer systems is essential
- Transaction standards – typically X12

Revenue Cycle Management - EA Application Layer



Revenue cycle functionality exemplifies why HIS systems are so complex!

RCM Technology Success Factors

- Workflow optimization and alignment
- Understanding of payer requirements like:
 - Care performance requirements
 - Coverage definition
 - Regulatory compliance
 - Risk management
- Data capture and governance
- Claim management – like coding
- Denial management
- Patient engagement
- Analytics and reporting

If Time
Revenue Cycle Executive Interview
Jamie Watson
VP Revenue Cycle
Boston Medical Center

In CS581 Blackboard Course Site.

Assignment 3 – Business Function Requirements

- Select one of the following two areas of HIS/EHR business functionality:
 - Population Health
- or
 - Revenue Cycle Management.
- Locate an article or publication published since January 1, 2024, that elaborates on the use or need for this area of functionality.
- Write a 4-8 paragraph requirements document.
- The requirements document:
 - Should elaborate on that area of need based on the article or publication, the online lecture, the textbook, and the VWC Davies Award Application;
 - Describe 5 to 8 key functional requirements in this area to meet the need; and
 - Discuss the reasons these 5 to 8 requirements are important.
- The requirements document should be written so it can be read “stand-alone.” Diagrams, bullet points, etc. are all fine.
- Submit your document in MS Word, Google Docs or Adobe PDF format on the course site.

Assignment 3 – Business Function Requirements

Learning Objective	Points
• Understanding of the requirements for either population health or revenue cycle management • Elaboration on that requirement based on the course lectures, the textbook, the Live Classroom, and an article or publication published since January 1, 2024 • Relationship of the requirements to VWC	45 Points
• Identification of key functions to address the selected area of business requirement as described in the lectures, the textbook, the VWC case, the Live Classroom, and the article. • How these functions capabilities address the requirements	45 Points
Clarity and quality of the material, citations, use of AI documented	10 points
Total	100 Points